

Decision Theory

Solution Sheet 1 and 2

Sheet 1: ML Estimation and Limiting Distributions

Problem 1: Poisson Distribution

Let $X_i \sim \text{Poi}(\theta)$. **1. Find the Maximum Likelihood (ML) estimator.**

$$L(\theta|x) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} = \frac{e^{-n\theta} \theta^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

Log-Likelihood:

$$\ln L(\theta|x) = -n\theta + \sum_{i=1}^n x_i \ln \theta - \sum_{i=1}^n \ln(x_i!)$$

FOC: $\frac{\partial \ln L}{\partial \theta} = -n + \frac{1}{\theta} \sum_{i=1}^n x_i = 0 \implies \hat{\theta}_{ML} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$. SOC: $\frac{\partial^2 \ln L}{\partial \theta^2} = -\frac{1}{\theta^2} \sum_{i=1}^n x_i < 0 \implies \text{Maximum}$.

2. Limiting Distribution of $\hat{\theta}_{ML}$: $E(\hat{\theta}_{ML}) = \theta$ and $\text{Var}(\hat{\theta}_{ML}) = \frac{\theta}{n}$. By the CLT:

$$\sqrt{n}(\bar{x} - \theta) \xrightarrow{d} N(0, \theta) \implies \hat{\theta}_{ML} \sim N\left(\theta, \frac{\theta}{n}\right)$$

3. Method of Moments: $E(X^2) = \text{Var}(X) + (E(X))^2 = \theta + \theta^2$. Setting the theoretical moment equal to the sample moment:

$$\overline{x^2} = \hat{\theta} + \hat{\theta}^2 \implies \hat{\theta}^2 + \hat{\theta} - \overline{x^2} = 0$$

Solving for $\hat{\theta}_{MM}$ (keeping the positive root):

$$\hat{\theta}_{MM} = -\frac{1}{2} + \sqrt{\frac{1}{4} + \overline{x^2}}$$

Problem 2: Posterior Kernel

Let $X_i \sim \text{Poi}(\theta)$. Prior: $f(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} I_{(0,\infty)}(\theta)$. **1. Determine the posterior kernel:**

$$f(\theta|x) \propto f(x|\theta)f(\theta) = \left(e^{-n\theta} \theta^{\sum x_i}\right) \left(\theta^{\alpha-1} e^{-\beta\theta}\right) = \theta^{\sum x_i + \alpha - 1} e^{-\theta(n+\beta)}$$

This is the kernel of a Gamma distribution: $\theta|x \sim \text{Gamma}(\sum x_i + \alpha, n + \beta)$.

2. Expected value of the posterior: For $Y \sim \text{Gamma}(a, b)$, $E(Y) = a/b$. Therefore:

$$E(\theta|X) = \frac{\sum_{i=1}^n x_i + \alpha}{n + \beta}$$

Problem 3: Hypothesis Testing

$X_i \sim N(\mu, \sigma^2)$ with $\sigma^2 = 0.01$. $H_0 : \mu \leq 15$. Critical region $C = \{\bar{x} : t(\bar{x}) = \frac{\bar{x}-15}{\sqrt{0.01/n}} > 1.645\}$.

1. Power function and Size: Asymptotic distribution: $\bar{x} \sim N(\mu, \frac{0.01}{n})$.

$$\Pi(\mu) = P(t(\bar{x}) > 1.645 | \mu) = 1 - \Phi\left(1.645 - \frac{\mu - 15}{\sqrt{0.01/n}}\right)$$

Size $\alpha = \sup_{\mu \leq 15} \Pi(\mu)$. The maximum occurs at $\mu = 15$:

$$\alpha = 1 - \Phi(1.645) = 0.05$$

2. Unbiasedness and Consistency: Unbiased because $\inf_{\mu > 15} \Pi(\mu) \geq 0.05$. Consistent because $\lim_{n \rightarrow \infty} \Pi(\mu) = 1$ for $\mu > 15$.

Sheet 2: Expected Loss and Decision Trees

Problem 1: Crypto Crash

Prior: $\pi(\theta_1) = 0.6$ (Crash), $\pi(\theta_2) = 0.4$ (No Crash). Losses: $L(\theta_1, d_1) = 1000$, $L(\theta_2, d_1) = 1000$, $L(\theta_1, d_2) = -40$, $L(\theta_2, d_2) = -40$.

$$R(\pi, d_1) = 0.6(-1000) + 0.4(1000) = -200$$

$$R(\pi, d_2) = 0.6(-40) + 0.4(-40) = -40$$

Problem 2: Heating Problem

Day 1 expected cost: Without heating: $0.7(0) + 0.3(20) = 6\text{€}$ With heating: $0.9(0) + 0.1(20) + 6 = 8\text{€}$ (Prefer not to change) **Day 2 expected cost (Cumulative):** Without heating: $0.7^3(0) + 0.7^2(0.3)(20) + 0.7(0.3)(40) + 0.3(60) = 29.34\text{€}$ With heating: $0.9^3(0) + 0.9(0.1)(20) + 0.9(0.1)(40) + 0.1(60) + 18 = 29.22\text{€}$ We should turn up the heat for the 2-day period.

Problem 3: Quadratic Loss

$$L = c(\hat{\mu} - \mu)^2.$$

$$R(\theta, d) = E[c(\hat{\mu} - \mu)^2] = c \cdot \text{MSE}(\hat{\mu})$$

Since sample average is unbiased, MSE is just the variance: $R = c \cdot \text{Var}(\hat{\mu}) = c \frac{\sigma^2}{n}$.